Brian Ngo

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Education

West Virginia University — Morgantown, WV

Bachelor of Science in Electrical and Computer Engineering | GPA: 4.0 — *Expected May 2028*

Coursework: Digital Logic Design, Digital Electronics, Analog Electronics, Microprocessor Systems, Electrical Circuits, Signals and Systems, Software Engineering

Experience

West Virginia University, Lane Department of CSEE — Aug 2026 – Present

Teaching Assistant, Digital Logic Design CPE-271 — *Morgantown, WV*

- Instruct digital logic lab section with 20 students, lecturing on HDL design (**VHDL**), **FPGAs**, functional simulation, combinational logic design sequential logic design, and digital hardware modules
- Audited lab procedures, documenting 10+ issues and improvements that can be applied next semester
- Run class review sessions for exams, going over foundational digital logic concepts: number systems, logic gates, CMOS, Boolean Algebra, K-maps, Muxes, Timing, Delay, and FSMs

Cyber-Resilience Resource Center — May 2026 – Aug 2026

Software Engineer Intern — *Morgantown, WV*

- Modeled attack surface traversal with **Dijkstra's shortest-path** over a **MITRE ATT&CK** technique graph
- Computed and displayed residual risk scores by combining **CVSS** severity data with existing control coverage
- Automated functional verification with **TypeScript** test suite and mock data scripts

Agile5 Technologies, Inc. — May 2025 – April 2026

Software Development Intern — *Fairmont, WV*

- Built an audio signal processing and transcription pipeline in **Python** with **FFmpeg** and the **Whisper API** on **Databricks**
- Queried 400,000+ federal records in **SQL** and shipped 7 full-stack features backed by **Databricks**

Projects

8-Bit CPU

- Implemented a simple 8-bit CPU with three op-codes and 32 bytes of RAM
- Created architecture diagrams mapping datapath, control signals, and component interconnect
- Verified the design against requirements with directed vector waveform tests in **Quartus Prime**, and ran the design on an **Intel MAX 10 FPGA**

Custom Mechanical Keyboard PCB

- Captured the switch matrix, microcontroller, and **USB-C** power and signal schematic in **KiCad**, selecting parts against electrical and footprint requirements
- Placed components and routed the board, including the **USB differential pair**, clearing electrical rule and design rule checks

CMOS Logic Design and Noise Margin Analysis

- Built static and clocked **CMOS** realizations of A+BC in **LTspice** with precharge, footer, and inverter stages
- Extracted VIH, VIL, VOH, and VOL from the transfer curve and sized W/L to balance noise margins within 0.15 V

Technical Skills

Programming & HDL: VHDL, Python, C/C++, MATLAB, ARM Assembly, TypeScript, SQL, Java

Tools: Quartus Prime, Altera Questa, DE10-Lite Intel FPGA, LTspice, QSpice, KiCad, Git, Jira, Linux, Databricks

Lab & Testing: Oscilloscope, multimeter, function generator, bench-top DC supply